IAT New Test 14

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

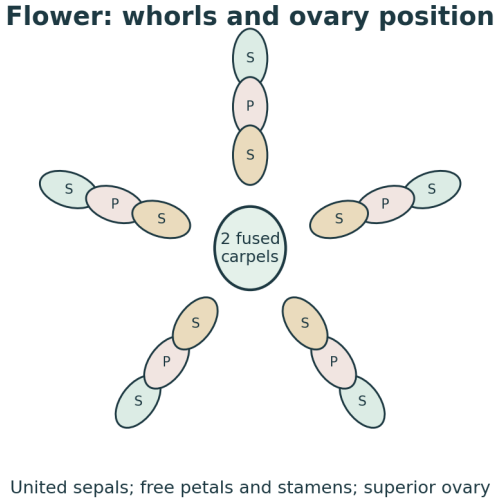
Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A moss sporophyte remains attached to its leafy gametophyte. If the gametophyte is deprived of photosynthate while the capsule is still developing, which outcome is most directly predicted?
A) Capsule maturation may fail because sporophyte nutrition depends on the gametophyte
B) The sporophyte immediately becomes a free-living fern
C) The capsule changes its ploidy from diploid to haploid
D) Every spore becomes a gamete without meiosis

Q2 A floral diagram has five united sepals, five free petals, five stamens and a superior ovary of two fused carpels. Which formula best matches it? Here brackets denote fusion.



A) K5 C(5) A5 G(2) inferior
B) K(5) C5 A5 G(2) superior
C) K(5) C5 A(5) G2 superior
D) K5 C5 A5 G2 inferior

Q3 Cells exposed to a spindle poison replicate DNA normally but fail to attach some kinetochores. Which stage is expected to pause before sister chromatids separate?
A) G1 before S phase
B) Telophase after cytokinesis
C) Metaphase-to-anaphase transition
D) Prophase before chromosome condensation

Q4 In bright light with closed stomata, a C3 leaf shows net CO_2 uptake falling sharply, while a C4 leaf retains appreciable uptake at the same temperature. Which explanation best fits the contrast?
A) C4 plants never use RuBisCO
B) C3 plants cannot make ATP under bright light
C) C4 mesophyll lacks any oxygen
D) C4 CO_2 concentration in bundle-sheath cells suppresses RuBisCO oxygenation

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Q5 A heart pumps 5 L min^{-1} into a systemic circuit whose effective mean pressure drop is 100 mm Hg. If flow stays 5 L min^{-1} but vascular resistance falls by 20%, what pressure drop is required in the same linear-flow approximation?
A) 80 mm Hg
B) 100 mm Hg
C) 120 mm Hg
D) 64 mm Hg

Q6 After fertilisation, an ovule's integuments remain intact and the ovary wall expands. Which mature structures derive from these two tissues, in that order?
A) Endosperm and embryo
B) Seed coat and fruit wall
C) Embryo and seed coat
D) Fruit wall and endosperm

Q7 Two unaffected parents have a child with an autosomal recessive disorder. A second child is unaffected. If a new point mutation creates the same disease allele in one parent's germ line, which claim about the unaffected second child remains justified from the observed pedigree alone?
A) The child must be homozygous normal
B) The child must be a carrier
C) The child could be either homozygous normal or a carrier
D) The child's somatic cells must contain the new germ-line mutation

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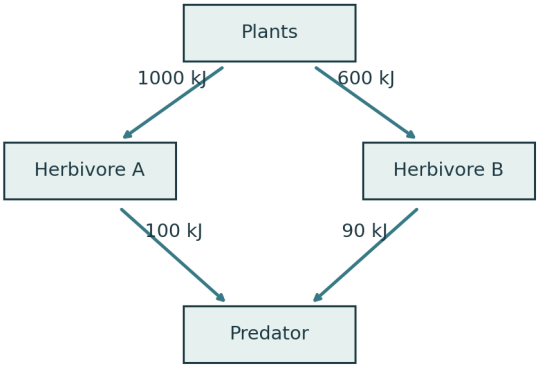
Q8 A virus infects host cells, and extracellular viral particles are neutralised by serum antibodies while already infected cells remain. Which additional response most directly removes those infected cells?
A) Plasma cells releasing more neutralising antibody
B) Complement acting only on free virions
C) Memory B cells without effector activation
D) Cytotoxic T cells recognising antigen displayed by infected cells

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Q9 A plasmid has intact marker S and a cloning site inside marker R. After ligation, colonies grow on S-antibiotic; a colony fails on R-antibiotic. Which control most directly tests whether failure is due to insertion at R rather than a nonrecombinant vector's ordinary phenotype?
A) Transform and plate an uncut parent vector on both antibiotics
B) Repeat only the same S plate without a vector control
C) Measure colony diameter without changing selection
D) Sequence an unrelated chromosomal gene

Q10 In the pictured food web, plants supply 1000 kJ to herbivore A and 600 kJ to herbivore B; a predator assimilates 100 kJ from A and 90 kJ from B. What fraction of the herbivores' combined incoming energy reaches that predator by those routes?

Energy transfers in the food web



A) 19/100
B) 19/160
C) 1/10
D) 3/20

Q11 At a locus, measured allele frequencies are p=0.7 and q=0.3. Observed genotype frequencies are AA=0.60, Aa=0.20, aa=0.20. Which conclusion is supported?
A) The allele frequencies cannot be 0.7 and 0.3
B) The genotype frequencies exactly match Hardy–Weinberg proportions
C) The genotype frequencies depart from Hardy–Weinberg proportions despite matching the stated allele frequencies
D) Heterozygotes must have frequency 0.49 at equilibrium

Q12 After a pesticide is applied, a rare heritable resistant insect type becomes common. Which additional finding best distinguishes natural selection from an induced, directed resistance mutation in every survivor?
A) All survivors are identical in appearance
B) Resistance occurs only after exposure
C) Pesticide concentration was high
D) Resistance was detected in archived pre-exposure specimens

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Q13 At fixed tissue oxygen demand, a blood sample with lower pH releases more O_2 at the same partial pressure than a control. Which shift of the haemoglobin dissociation curve is consistent with that observation?
A) Left, higher affinity
B) No shift, unchanged affinity
C) Only a vertical shift in maximum haemoglobin amount
D) Right, lower affinity

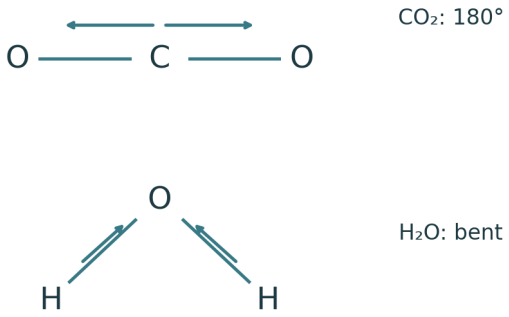
Q14 A disaccharide is hydrolysed to two monosaccharides without changing their carbon skeletons. Which bond was cleaved directly?
A) A glycosidic bond between sugar residues
B) A peptide bond between amino acids
C) A phosphodiester bond in nucleic acid
D) A disulfide bond between cysteines

Q15 In two habitat surveys the same sampling method finds 10 species in area A and 40 species in area 16A. If species number follows S=cA^z over this scale, what is z?
A) 1/4
B) 1/2
C) 1
D) 2

Chemistry

Q16 The diagram places two identical polar C–O bonds symmetrically at 180° in CO_2, and two polar O–H bonds at an angle in H_2O. Which molecule has zero net electric dipole in its shown geometry?

Bond arrangement and bond dipoles



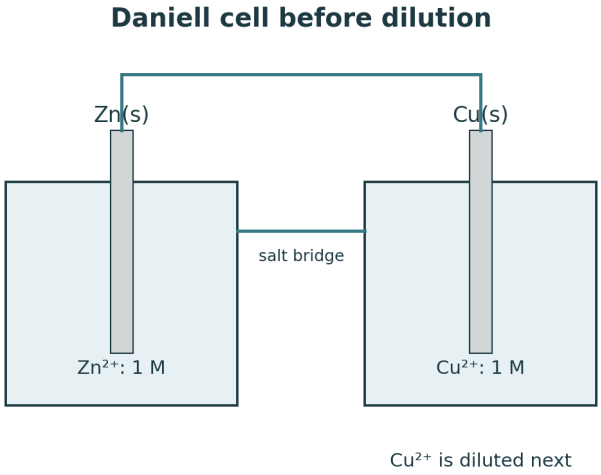
A) CO_2 only
B) H_2O only
C) Neither
D) Both

Q17 A complex $[M(NH_3)_4Cl_2]Cl$ contains a mononuclear metal M; all ligands inside brackets are monodentate and NH_3 neutral. What are the coordination number and oxidation state of M?
A) 6 and +3
B) 4 and +2
C) 6 and +2
D) 7 and +3

Q18 For a reaction with $\Delta H = -20$ kJ mol^{-1} and $\Delta S = -50$ J mol^{-1} K^{-1}, at what temperature range is ΔG negative under the constant-ΔH,ΔS approximation?
A) Above 400 K
B) Below 400 K
C) At all positive temperatures
D) At no positive temperature

Q19 Under base-catalysed aldol conditions, which carbonyl compound can form an enolate from its own α-hydrogen and undergo self-aldol addition?
A) Benzaldehyde
B) Formaldehyde
C) Propanal
D) Benzophenone

Q20 A Daniell cell is built with $Zn(s)|Zn^{2+}(1\ M)$ and $Cu^{2+}(1\ M)|Cu(s)$. Immediately after a small amount of Cu^{2+} solution is diluted without changing Zn^{2+} concentration, how does the reaction quotient $Q=[Zn^{2+}]/[Cu^{2+}]$ change and what is the direction of E relative to its earlier value?



A) Q falls, E rises
B) Q rises, E rises
C) Q falls, E falls
D) Q rises, E falls

Q21 An unknown amine gives a positive carbylamine test with $CHCl_3$ and alcoholic KOH. A separate sample gives a cold, relatively stable diazonium salt with $NaNO_2$/HCl at 0–5°C, and this couples with alkaline phenol. Which class best fits both observations?
A) Primary aromatic amine
B) Primary aliphatic amine
C) Secondary aromatic amine
D) Tertiary aromatic amine

Q22 In acidic solution, $3MnO_4^{2-}+4H^+\rightarrow 2MnO_4^-+MnO_2+2H_2O$ is proposed. Is this a disproportionation of Mn(VI), and which products hold its oxidation and reduction outcomes?
A) No, Mn remains +6 in all species
B) Yes, Mn(VII) in MnO_4^- and Mn(IV) in MnO_2
C) Yes, Mn(V) and Mn(VI) are products
D) No, oxygen alone undergoes redox

Q23 For a reaction with activation energy 50 kJ mol^{-1}, estimate $\ln(k_2/k_1)$ when temperature rises from 300 K to 330 K; use R=8.3 J mol^{-1} K^{-1} and constant activation energy.
A) −1.83
B) 0.18
C) 1.83
D) 18.3

Q24 Acetyl chloride is added slowly to excess water. Which principal organic product forms after hydrolysis?
A) Ethanol
B) Ethanal
C) Acetic anhydride
D) Ethanoic acid

Q25 Two compounds have formula C_2H_6O: one is ethanol and the other dimethyl ether. Which relationship describes them?
A) Functional isomers
B) Geometrical isomers
C) Optical isomers
D) Conformers of one molecule

Q26 At the same molality, ideal dilute aqueous glucose and ideal fully dissociated NaCl solutions are compared. What is the ratio of the magnitude of NaCl freezing-point depression to glucose depression?
A) 1
B) 2
C) 3
D) 1/2

Q27 A two-step preparation is drawn: A+2B→C, then C+B→D. Initially there are 3 mol A and 8 mol B, and the two steps can be allocated optimally. What is the maximum D produced?

Two-step preparation



Initial amounts: A = 3 mol; B = 8 mol

A) 3 mol
B) 2 mol
C) 8/3 mol
D) 8 mol

Q28 An element has successive ionisation energies I_1=580, I_2=1817, I_3=2745 and I_4=11578 kJ mol^{-1}. Which number of valence electrons and main-group number are consistent with the sharp jump?
A) One; group 1
B) Three; group 13
C) Two; group 2
D) Four; group 14

Q29 A DNA duplex region has 30 adenine bases and 20 guanine bases across both strands. How many thymine and cytosine bases, respectively, must it contain under ordinary Watson–Crick pairing?
A) 20 and 30
B) 30 and 30
C) 30 and 20
D) 20 and 20

Q30 A buffer contains 0.20 mol CH_3COOH and 0.10 mol CH_3COO^- in 1 L. Add 0.05 mol strong base with negligible volume change. What is the new acid-to-base mole ratio before using any logarithmic pH formula?
A) 4:1
B) 2:1
C) 3:1
D) 1:1

Mathematics

Q31 The circle $(x-2)^2+(y+1)^2=9$ touches which of the following horizontal lines?
A) $y=2$
B) $y=0$
C) $y=-1$
D) $y=4$

Q32 On the closed interval $[-1,2]$, what is the maximum of $f(x)=x^3-3x+1$?
A) 5
B) 1
C) 2
D) 3

Q33 A curve is given implicitly by $x^2y+y^3=2$. At its point $(1,1)$, what is the tangent slope?
A) -2
B) $1/2$
C) 0
D) $-1/2$

Q34 Evaluate $\int_0^1 dx/[(x+1)^2(x+2)]$.
A) $\ln(4/3)$
B) $1/2-\ln(4/3)$
C) $1/2+\ln(4/3)$
D) $1/2$

Q35 For $f(x)=x^2$ on $x\geq 0$, which inverse is valid as a function with codomain $[0,\infty)$?
A) $-\sqrt{x}$
B) $\pm\sqrt{x}$
C) $\sqrt{x}$
D) $1/x^2$

Q36 Vectors $a=(1,1,0)$ and $b=(1,-1,2)$ are viewed by an observer on the positive z-axis looking toward the origin. Which pair gives $a\times b$ and whether its z-component points into or out of the screen?
A) $(-2,2,2)$, out of screen
B) $(2,-2,2)$, out of screen
C) $(2,-2,0)$, in the screen
D) $(2,-2,-2)$, into the screen

Q37 Let $f:\mathbb{R}\to\mathbb{R}$ be $f(x)=x^3+x$. Which property guarantees it has an inverse defined on all $\mathbb{R}$?
A) It is continuous, strictly increasing and unbounded in both directions
B) It is even
C) It has derivative zero everywhere
D) It is periodic

Q38 Two different tickets are drawn without replacement from five numbered 1 to 5, with unordered pairs equally likely. Given that their product is even, what is the probability both tickets are even?
A) 1/10
B) 1/7
C) 2/5
D) 1/2

Q39 How many nonempty multisets of size 4 can be chosen from 3 available types if supply is unlimited?
A) 12
B) 10
C) 15
D) 14

Q40 Let p be a positive integer. In the expansion of $(x^p+x^{-2})^6$, the coefficient of the term independent of x is 15. What is the sum of all possible p?
A) 1
B) 4
C) 6
D) 5

Q41 For which real x does the infinite series $\sum_{n=1}^{\infty} (x-2)^n/n$ converge?
A) [1,3)
B) (1,3)
C) [1,3]
D) (−1,1)

Q42 Polynomials x^2-3x+2 and x^2-kx+4 share one real root. What is the sum of all real possible k?
A) 5
B) 9
C) 7
D) 6

Q43 A data set has variance 9. If every value is replaced by 2x+5, what is the new variance?
A) 6
B) 18
C) 36
D) 49

Q44 A one-parameter family of curves is $y=Cx^2$ for $x>0$. Eliminate C to form its differential equation.
A) $y'=Cx$
B) $xy'=y$
C) $y'=2y/x$
D) $y'=2x/y$

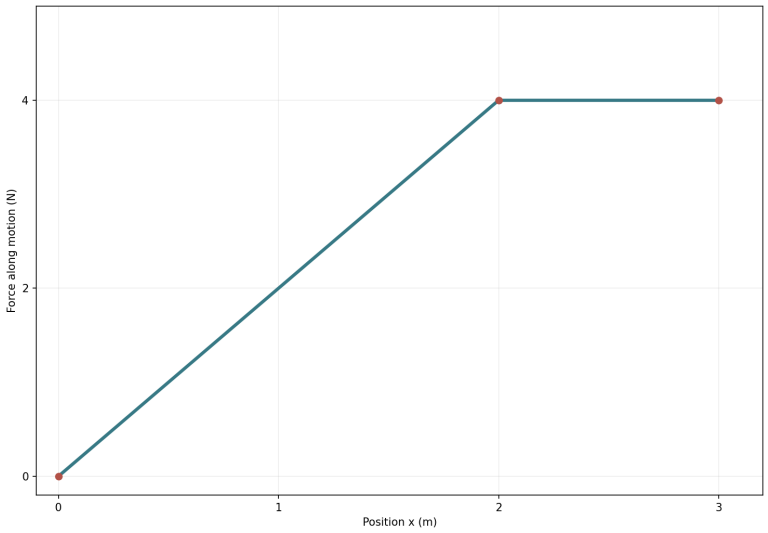
Q45 Evaluate $\lim_{x\to 0} (e^{2x}-1-2x)/x^2$.
A) 2
B) 1
C) 4
D) 0

Physics

Q46 A ball is thrown vertically upward at 20 m s^{-1} from a balcony; take g=10 m s^{-2}. At what time is its height above the balcony greatest?
A) 1 s
B) 2 s
C) 3 s
D) 4 s

Q47 An elevator accelerates upward at 2 m s^{-2}. A 3 kg mass hangs stationary relative to its cabin from a light cord. What tension does the cord exert? Use g=10 m s^{-2}.
A) 24 N
B) 30 N
C) 36 N
D) 42 N

Q48 A 2 kg particle enters x=0 at 2 m s^{-1}. The pictured force along its motion is F(x)=2x N from 0 to 2 m, then 4 N from 2 to 3 m. With no other work, what speed does it have at x=3 m?



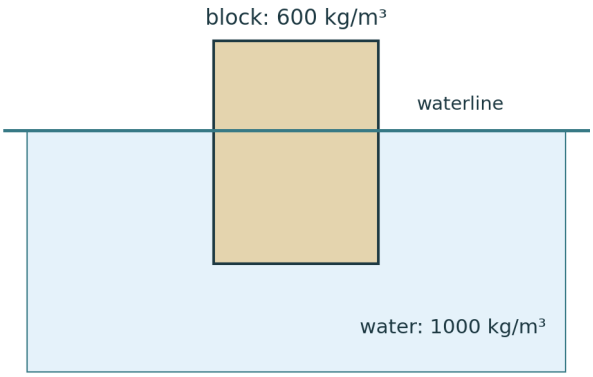
A) 2 m s^{-1}
B) √8 m s^{-1}
C) 4 m s^{-1}
D) 2√3 m s^{-1}

Q49 A uniform solid disk rolls without slipping down from rest through vertical height h. What fraction of its final kinetic energy is rotational about its centre?
A) 1/3
B) 1/4
C) 1/2
D) 2/3

Q50 A planet has twice Earth's mass and twice Earth's radius. What is its surface escape speed divided by Earth's, neglecting rotation?
A) 1/√2
B) 1
C) √2
D) 2

Q51 A uniform block of density 600 kg m⁻³ floats upright in water of density 1000 kg m⁻³. What fraction of its height is submerged at equilibrium?

Block floating at equilibrium



A) 0.4
B) 0.5
C) 0.6
D) 1.0

Q52 A gas absorbs 50 J heat and does 20 J work on surroundings. Under $\Delta U = Q - W$, what is its internal-energy change?
A) −70 J
B) −30 J
C) 70 J
D) 30 J

Q53 Two identical positive charges +q are fixed at x=−a and x=+a. At which finite x on their joining axis is the net electric field zero?

Equal positive charges on one axis



A) x=0
B) x=−2a
C) x=2a
D) No finite point

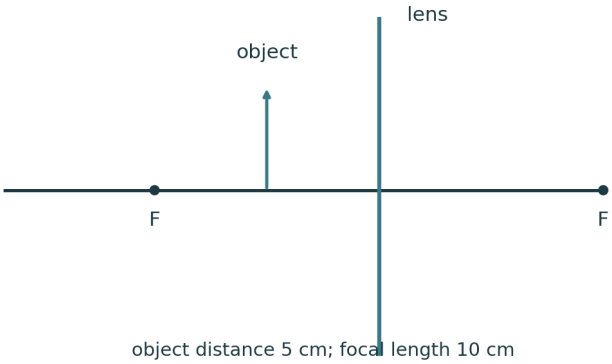
Q54 A capacitor is charged and then disconnected from its battery. A dielectric of relative permittivity 4 completely fills the gap. What is final-to-initial electrostatic energy ratio?
A) 4
B) 1/4
C) 1/16
D) 16

Q55 A charge +q moves at v=2 m s^{-1} along +x in a magnetic field B=3 T along +y. What is magnetic-force direction and magnitude, for q=1 C?
A) −z, 6 N
B) +z, 3 N
C) +z, 6 N
D) −y, 6 N

Q56 An ideal inductor of 2 H carries 3 A. What energy is stored in its magnetic field?
A) 3 J
B) 6 J
C) 12 J
D) 9 J

Q57 A convex lens with f=10 cm has an object 5 cm in front of it. Which image is formed, using Cartesian signs with light left to right?

Converging lens and near object



A) Virtual upright on object side, 10 cm from lens
B) Real inverted on far side, 10 cm
C) Virtual inverted on far side, 5 cm
D) Real upright on object side, 5 cm

Q58 In a single-slit Fraunhofer setup, slit width is halved while wavelength and screen distance remain fixed. What happens to the central maximum's linear width?
A) Halves
B) Doubles
C) Quadruples
D) Unchanged

Q59 Which spectral feature distinguishes a hot dilute atomic gas from a hot dense source viewed through a dispersing prism?
A) Continuous spectrum in all cases
B) No radiation from dilute gas
C) Bright discrete emission lines at characteristic wavelengths
D) Only a single universal line for all elements

Q60 A detector counts 800 events per minute from a radioactive source plus constant 100-per-minute background. Two half-lives later, what total count rate should it read, assuming detector efficiency unchanged?
A) 175
B) 200
C) 300
D) 275

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A bryophyte sporophyte dependence

Answer: (A) The developing capsule may lack supply.

Why: In bryophytes the sporophyte is attached to and nutritionally dependent on the dominant gametophyte, so removing that supply threatens its growth. The capsule still contains diploid sporogenous tissue until meiosis produces haploid spores; starvation does not rewrite that life-cycle sequence.

Why not the others: (B) changes the taxon; (C) confuses nutritional state with chromosome number; (D) replaces meiotic spore formation with gamete production.

Key point: Distinguish dependence from ploidy.

Q2 / B floral formula inference

Answer: (B) K(5) C5 A5 G(2), superior.

Why: The calyx is fused, petals and stamens are free, and the bicarpellary gynoecium is syncarpous. The diagram places other whorls below the ovary, supporting the superior assignment; no fusion of stamens is indicated.

Why not the others: (A) swaps sepal/petal fusion and ovary position; (C) invents stamen fusion and free carpels; (D) loses both fusions and reverses ovary position.

Key point: Translate each depicted whorl independently.

Q3 / C mitotic checkpoint

Answer: (C) The spindle checkpoint delays anaphase.

Why: Unattached kinetochores signal that chromosome segregation is not yet safe, preventing the transition that separates sister chromatids. DNA replication already occurred, so an S-phase arrest does not explain the observed attachment defect.

Why not the others: (A) is earlier than replication; (B) is after segregation; (D) concerns condensation rather than attachment.

Key point: A checkpoint responds to a specific unfinished process.

Q4 / D photorespiration contrast

Answer: (D) Spatial concentration suppresses photorespiration.

Why: Closing stomata lowers internal CO_2 and favours RuBisCO oxygenation in C3 leaves, reducing net fixation. C4 fixation and transfer elevate CO_2 near bundle-sheath RuBisCO, so the same external limitation has a smaller photorespiratory penalty.

Why not the others: (A) C4 bundle sheaths use RuBisCO; (B) light reactions persist; (C) oxygen is present in C4 tissues.

Key point: Compare local RuBisCO environments.

Q5 / A cardiac output and pressure

Answer: (A) 80 mm Hg.

Given: Pressure drop is flow times effective resistance in the stipulated model.

Formula/Principle: $\Delta P = QR$; Q unchanged, $R_{new} = 0.8 R_{old}$.

Step 1: $\Delta P_{new} = 0.8 \times 100 = 80$ mm Hg.

Why: Holding cardiac output fixed means the pressure gradient scales with the resistance, not with stroke volume or heart rate separately. This is a simplified mean-flow model; it does not assert that every instantaneous arterial pressure follows the same ratio.

Why not the others: (B) ignores changed R; (C) raises instead of lowers it; (D) incorrectly squares the factor.

Key point: Keep the held quantity explicit.

Q6 / B seed and fruit development

Answer: (B) Seed coat, fruit wall.

Why: The integuments surrounding the ovule become the protective seed coat, while the ovary wall develops into the fruit wall or pericarp. Endosperm and embryo arise from fertilisation products, not directly from either of the named maternal protective layers.

Why not the others: (A) names fertilisation products; (C) assigns embryo to integuments; (D) reverses tissue origin and inserts endosperm.

Key point: Trace maternal tissues and fusion products separately.

Q7 / C mutation and disorder

Answer: (C) The unaffected genotype is not uniquely fixed.

Why: Under recessive inheritance, unaffected status permits both AA and Aa, even when parental transmission can be complicated by a new germ-line mutation. A mutation arising in a parent's germ line is not automatically present in that parent's somatic cells or in every sibling.

Why not the others: (A) excludes carriers without evidence; (B) treats possible carrier status as certain; (D) forces inheritance of an event from another gamete.

Key point: Distinguish phenotype from an unobserved genotype.

Q8 / D antibody versus cellular defence

Answer: (D) Cytotoxic T-cell effector response.

Why: Antibody can block extracellular entry but cannot by itself reach viral proteins inside an infected cell. Cytotoxic T cells recognise peptides presented on infected-cell MHC I and can kill the cellular reservoir, addressing the residual infection described.

Why not the others: (A) acts chiefly on extracellular particles; (B) does not target infected intracellular reservoirs as stated; (C) is a memory state rather than immediate cellular killing.

Key point: Match the pathogen's location to defence mechanism.

Q9 / A recombinant selection control

Answer: (A) Compare intact parent-vector phenotype.

Why: An uncut plasmid should retain both markers, establishing that R-antibiotic growth is possible for the parent vector under the same conditions. If the recombinant colony alone loses R growth while retaining S, insertional disruption becomes more credible than a baseline selection failure.

Why not the others: (B) does not test R baseline; (C) does not assay marker function; (D) misses the plasmid locus.

Key point: A control must preserve the causal feature being tested.

Q10 / B food-web energy accounting

Answer: (B) 19/160.

Given: Predator receives 190 kJ; the two herbivore inputs sum to 1600 kJ.

Formula/Principle: Fraction is assimilated output divided by combined incoming energy.

Step 1: 190/1600=19/160.

Why: Both prey pathways contribute to the predator, so the numerator must include their separate transfers; the denominator is the sum of plant-to-herbivore supplies specified in the diagram. This fraction is not a universal trophic-transfer efficiency because unshown losses and other pathways are outside the stated budget.

Why not the others: (A) uses 1000 alone; (C) imposes a generic 10% rule; (D) omits one route.

Key point: Add route-specific flows before forming a ratio.

Q11 / C Hardy-Weinberg assumption

Answer: (C) Frequencies deviate from equilibrium.

Given: Observed A allele frequency=0.60+0.20/2=0.70.

Formula/Principle: Equilibrium expected p^2,2pq,q^2=0.49,0.42,0.09.

Step 1: Compare expected to 0.60,0.20,0.20; all differ.

Why: The allele frequencies can be correct while genotype proportions depart from random-mating equilibrium. The result alone does not identify which assumption failed, such as selection, structure or nonrandom mating.

Why not the others: (A) ignores observed allele calculation; (B) overlooks expected 0.42 heterozygotes; (D) confuses p^2 with 2pq.

Key point: Test genotype proportions, not just p+q=1.

Q12 / D evolutionary selection evidence

Answer: (D) Earlier occurrence supports selection.

Why: Finding the resistance phenotype before exposure establishes that a heritable variant existed for the pesticide to select. The observed increase after exposure can then be explained by differential survival and reproduction without assuming the chemical produced the needed mutation on demand.

Why not the others: (A) appearance may hide genotype; (B) would not establish pre-existing variation; (C) says little about origin.

Key point: Temporal evidence separates variation from selection.

Q13 / D gas transport and dissociation

Answer: (D) Right shift and greater unloading.

Why: Higher H^+ concentration stabilises lower-affinity haemoglobin states, so the same oxygen partial pressure gives lower saturation and more release to tissues. The curve shift concerns binding equilibrium, not a change in the number of haemoglobin molecules in the sample.

Why not the others: (A) predicts less unloading; (B) contradicts measured release; (C) changes capacity rather than affinity.

Key point: Read saturation at fixed partial pressure.

Q14 / A biomolecule linkage

Answer: (A) Glycosidic linkage.

Why: Condensation joins monosaccharides through an O-glycosidic linkage, and hydrolysis adds water across that linkage to release the original sugar units. The other named bonds connect different biomolecular building blocks and do not define a disaccharide.

Why not the others: (B) joins amino acids; (C) joins nucleotides; (D) stabilises proteins.

Key point: Identify linkage from monomer identity.

Q15 / B species-area relationship

Answer: (B) z=1/2.

Given: $S_2/S_1=4$ and $A_2/A_1=16$.

Formula/Principle: 4=16^z; take logarithms or compare powers.

Step 1: 16^(1/2)=4, giving z=1/2.

Why: The multiplicative constant c cancels between the two comparable surveys, allowing the exponent to be inferred without measuring absolute area or c. The fitted exponent describes only the stated scale and cannot by itself prove a mechanism of biodiversity gain.

Why not the others: (A) predicts ratio 2; (C) predicts 16; (D) predicts 256.

Key point: Form a ratio before taking logarithms.

Chemistry

Q16 / A bond polarity versus molecular dipole

Answer: (A) CO_2 only.

Why: Its equal bond-dipole vectors oppose along one line and cancel; the bent water vectors have a nonzero resultant toward oxygen. Polar bonds therefore do not guarantee a polar molecule unless their vector arrangement is considered.

Why not the others: (B) reverses the geometries; (C) ignores cancellation in linear CO_2; (D) ignores water's bend.

Key point: Sum bond dipoles as vectors.

Q17 / A coordination number and charge

Answer: (A) Six and +3.

Given: One external chloride makes the bracketed ion +1.

Formula/Principle: Four NH_3 plus two bound Cl give six donors; x−2=+1.

Step 1: x=+3, coordination number=6.

Why: The external counterion contributes to charge balance but is not coordinated, while both bracketed chlorides occupy donor positions. Confusing oxidation state with coordination number would misread two independent properties of the formula.

Why not the others: (B) omits bound chlorides; (C) misses counterion charge; (D) counts external chloride as ligand.

Key point: Count donors inside brackets and charges across brackets separately.

Q18 / B Gibbs-temperature boundary

Answer: (B) Below 400 K.

Why: Converting entropy to -0.050 kJ mol^{-1} K^{-1} gives $\Delta G = -20 + 0.050T$, which crosses zero at 400 K. At lower temperatures the favourable exothermic term dominates the unfavourable entropy decrease; above it the sign reverses.

Why not the others: (A) reverses inequality; (C) ignores entropy; (D) ignores enthalpy.

Key point: Convert units before solving the temperature boundary.

Q19 / C aldol eligibility

Answer: (C) Propanal.

Why: Propanal has an α-carbon bearing hydrogens next to its carbonyl, permitting enolate formation and attack on another propanal carbonyl. Benzaldehyde, formaldehyde and benzophenone lack an enolisable α-hydrogen for their own enolate under the stated mechanism.

Why not the others: (A) benzylic ring carbon is not an α-CH; (B) has no α-carbon; (D) has phenyl groups but no α-H.

Key point: Locate α-hydrogen on carbon next to carbonyl.

Q20 / D cell polarity from quotient

Answer: (D) Q rises and E falls.

Why: Zinc oxidation produces Zn^{2+} while copper reduction consumes Cu^{2+}; lowering the copper reactant concentration raises the quotient. By the Nernst form $E = E° - (RT/2F)\ln Q$, the increased Q lowers the potential at fixed temperature.

Why not the others: (A) reverses quotient; (B) reverses Nernst sign; (C) reverses both.

Key point: Write Q from the balanced cell reaction.

Q21 / A paired amine class and aromaticity tests

Answer: (A) Primary aromatic amine.

Why: A positive carbylamine test establishes a primary amine, while a diazonium salt that persists cold and couples with phenol points to an aryl primary amine. Primary aliphatic diazonium species decompose rapidly instead of giving the stated stable coupling response.

Why not the others: (B) can give carbylamine but fails the persistent diazonium observation; (C) is not primary and cannot satisfy both tests; (D) likewise fails the primary-amine requirement.

Key point: Combine independent class and aryl-group evidence.

Q22 / B disproportionation assessment

Answer: (B) Mn(VI) gives +7 and +4.

Why: Mn in MnO_4^{2-} is +6; in MnO_4^- it is +7 and in MnO_2 it is +4. One portion of the same initial oxidation state loses electrons while another gains them, and the stated stoichiometry balances the electron exchange.

Why not the others: (A) misassigns product states; (C) invents Mn(V); (D) overlooks manganese changes.

Key point: Determine oxidation numbers for each species.

Q23 / C Arrhenius temperature comparison

Answer: (C) About 1.83.

Given: $\ln(k_2/k_1)=Ea/R(1/T_1-1/T_2)$.

Formula/Principle: $50000/8.3 \times 30/(300\times330)$.

Step 1: $6024\times0.000303\approx1.83$.

Why: Increasing temperature makes 1/T fall, so the reciprocal difference is positive and rate constant increases. The logarithm compares rates through activation energy; it is not directly the fractional temperature increase.

Why not the others: (A) reverses temperatures; (B) misses a factor ten; (D) adds one.

Key point: Keep Ea and R in matching joule units.

Q24 / D acid derivative transformation

Answer: (D) Ethanoic acid.

Why: Water attacks the reactive acyl carbon and chloride leaves, converting CH_3COCl to CH_3COOH with HCl as a coproduct. Excess water drives hydrolysis rather than condensation to anhydride.

Why not the others: (A) needs reduction; (B) needs selective reduction; (C) forms under dehydrating acylation conditions.

Key point: Replace an acyl leaving group with hydroxyl under hydrolysis.

Q25 / A isomer classification

Answer: (A) Functional isomers.

Why: Their atoms count identically, but ethanol contains an O–H alcohol group whereas dimethyl ether has C–O–C connectivity. This difference in functional group is structural isomerism, not a different orientation of the same bond framework.

Why not the others: (B) needs restricted rotation geometry; (C) needs chirality; (D) preserves connectivity.

Key point: Same formula can encode different functional groups.

Q26 / B colligative-property comparison

Answer: (B) Two.

Given: Same Kf and molality; glucose i=1, NaCl i=2.

Formula/Principle: $\Delta Tf=iKfm$.

Step 1: Ratio=2/1=2.

Why: Freezing-point depression depends on the number of dissolved solute particles in the idealised comparison. The full-dissociation premise makes each NaCl formula unit contribute two ions, while glucose remains molecular.

Why not the others: (A) ignores dissociation; (C) counts three ions; (D) inverts ratio.

Key point: State the particle-count assumption.

Q27 / C limiting reagent in stages

Answer: (C) 8/3 mol.

Given: Each D requires one A and three B across both steps.

Formula/Principle: $Dmax=\min(nA,nB/3)$.

Step 1: $\min(3,8/3)=8/3$ mol; 16/3 mol B forms 8/3 mol C, leaving 8/3 mol B for the second step.

Why: The total net reaction $A+3B\rightarrow D$ makes B limiting at 8/3 mol, and the explicit allocation confirms this bound can be attained. Taking all three moles of A through step one would waste B needed for the second step.

Why not the others: (A) ignores B deficiency; (B) rounds down unnecessarily; (D) treats all B as product.

Key point: Sum stages and check an achievable allocation.

Q28 / B successive ionisation-energy jump and valence group

Answer: (B) Three valence electrons; group 13.

Why: The exceptionally large increase from the third to the fourth removal means the first three came from the outer shell, while the fourth would remove an inner-shell electron. A main-group atom with three outer electrons belongs to group 13 under modern numbering; the absolute values are not needed to identify that boundary.

Why not the others: (A) puts the jump after the first electron; (C) puts it after the second; (D) treats the costly core electron as valence.

Key point: Locate the largest successive-energy jump before assigning group.

Q29 / C nucleic-acid base pairing

Answer: (C) 30 thymine, 20 cytosine.

Why: Each A pairs with one T, and each G pairs with one C, so duplex totals satisfy A=T and G=C. The counts are across both strands; assigning complementary counts per individual strand is unnecessary.

Why not the others: (A) swaps complementary partners; (B) overcounts C; (D) undercounts T.

Key point: Use complementary base totals in a duplex.

Q30 / D buffer before and after neutralisation

Answer: (D) 1:1.

Given: OH^- consumes HA, producing A^-.

Formula/Principle: nHA=0.20−0.05=0.15; nA^-=0.10+0.05=0.15.

Step 1: Ratio 0.15:0.15=1:1.

Why: Neutralisation occurs before buffer equilibrium is recalculated, and both members change in opposite directions by the same stoichiometric amount. Using the original ratio would ignore the chemical reaction with added base.

Why not the others: (A) reverses addition; (B) keeps original ratio; (C) changes only the numerator incorrectly.

Key point: Update mole counts before Henderson–Hasselbalch.

Mathematics

Q31 / A circle tangency

Answer: (A) y=2.

Given: Centre (2,−1), radius 3.

Formula/Principle: A horizontal tangent is at centre y plus or minus radius.

Step 1: y=−1±3 gives 2 and −4; only 2 is listed.

Worked derivation:

$$d((2,-1), y=c) = |c+1| = 3 \Rightarrow c = 2 \text{ or } -4.$$

Why: Tangency means the perpendicular distance from centre to line equals radius. It is not enough that a line merely crosses the circle at two points.

Why not the others: (B) lies one unit away; (C) passes through centre; (D) lies five units away.

Key point: Use centre-to-line distance.

Q32 / D extreme value with endpoints

Answer: (D) 3.

Given: f′=3(x²−1); the interval endpoints are −1 and 2, with interior critical point 1.

Formula/Principle: Compare the function at every endpoint and interior critical point.

Step 1: f(−1)=3, f(1)=−1, f(2)=3, so the maximum is 3.

Worked derivation:

$$\max_{[-1,2]} f = \max\{f(-1), f(1), f(2)\} = \max\{3, -1, 3\} = 3.$$

Why: A closed interval requires both endpoints to be checked, and the interior stationary point is a minimum here. The maximum occurs at two different endpoints, but it has one numerical value.

Why not the others: (A) comes from overestimating an endpoint; (B) reads the constant term only; (C) omits the endpoint comparison.

Key point: Compare endpoints and critical points.

Q33 / D implicit tangent

Answer: (D) −1/2.

Given: The point satisfies 1²(1)+1³=2.

Formula/Principle: Implicit differentiation of x²y requires the product rule, and differentiation of y³ requires the chain rule.

Step 1: 2xy+(x²+3y²)y′=0; substitute (1,1) to obtain 2+4y′=0.

Worked derivation:

$$2xy + x^2y' + 3y^2y' = 0 \Rightarrow y' = -\frac{2xy}{x^2+3y^2} \Rightarrow y'(1,1) = -\frac{1}{2}.$$

Why: Both the coefficient of y′ in x²y and the derivative of y³ contribute to the denominator. Evaluating the derivative only after differentiating the full implicit relation preserves the tangent direction.

Why not the others: (A) omits the y′ terms from y³; (B) drops the negative sign; (C) treats y as constant.

Key point: Differentiate both products and powers of y.

Q34 / B repeated-pole partial fractions with limits

Answer: (B) 1/2−ln(4/3).

Given: The denominator has a repeated linear factor, so include both first and second powers of x+1.

Formula/Principle: 1/[(x+1)²(x+2)]=−1/(x+1)+1/(x+1)²+1/(x+2).

Step 1: Integrate to [−ln(x+1)−1/(x+1)+ln(x+2)]₀¹ and subtract endpoints.

Worked derivation:

$$I = \left[\ln\frac{x+2}{x+1} - \frac{1}{x+1}\right]_0^1 = \left(\ln\frac{3}{2} - \frac{1}{2}\right) - (\ln 2 - 1) = \frac{1}{2} - \ln\frac{4}{3}.$$

Why: A repeated pole requires a separate 1/(x+1)² term, whose integral is rational rather than logarithmic. Dropping that term gives the earlier simpler integral and the wrong value.

Why not the others: (A) omits the repeated-pole contribution; (C) reverses the endpoint subtraction; (D) cancels the logarithms incorrectly.

Key point: Include every power of a repeated linear factor.

Q35 / C inverse restriction

Answer: (C) √x.

Worked derivation:

$$y = x^2,\ x \geq 0 \Rightarrow x = \sqrt{y}, \quad f^{-1}(y) = \sqrt{y}.$$

Why: Restricting the original domain to nonnegative x makes squaring one-to-one and onto the specified nonnegative codomain. The inverse must return a single allowed nonnegative input for every output.

Why not the others: (A) returns excluded negative x; (B) is multivalued; (D) does not undo squaring.

Key point: Inverses depend on domain restriction.

Q36 / D cross-product orientation

Answer: (D) (2,−2,−2), into the screen.

Given: Positive z points toward the observer; the ordered product is a×b.

Formula/Principle: Use the determinant of the basis-vector array, retaining the negative sign on the j component.

Step 1: The three components are 1(2)−0(−1)=2, 0(1)−1(2)=−2 and 1(−1)−1(1)=−2.

Worked derivation:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 1 & -1 & 2 \end{vmatrix} = 2\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}, \quad (\mathbf{a} \times \mathbf{b})_z = -2 < 0.$$

Why: The negative z-component points away from an observer on the positive z-axis, so the perpendicular vector goes into the screen. Computing both in-plane components is necessary because the requested answer includes the complete cross product as well as the viewing orientation.

Why not the others: (A) reverses the ordered cross product; (B) loses the negative z sign; (C) discards the nonzero z component.

Key point: Keep determinant signs and viewing direction separate.

Q37 / A inverse existence

Answer: (A) Bijective onto $\mathbb{R}$.

Worked derivation:

$$f'(x) = 3x^2 + 1 > 0, \quad \lim_{x \to \pm\infty} f(x) = \pm\infty.$$

Why: Positive derivative gives injectivity, while continuity and opposite infinite limits give every real output by the intermediate value property. Both properties are necessary to define a unique inverse on the entire stated codomain.

Why not the others: (B) would collide x and −x; (C) is false; (D) would repeat outputs.

Key point: Check one-to-one and onto separately.

Q38 / B without-replacement conditional even product

Answer: (B) 1/7.

Given: There are three odd tickets and two even tickets.

Formula/Principle: Count pairs with even product by excluding the pairs of two odd tickets.

Step 1: Of C(5,2)=10 pairs, C(3,2)=3 have odd product; exactly C(2,2)=1 pair has both even.

Worked derivation:

$$P(\text{both even} \mid \text{product even}) = \frac{\binom{2}{2}}{\binom{5}{2} - \binom{3}{2}} = \frac{1}{10 - 3} = \frac{1}{7}.$$

Why: The observation rules out the three odd-odd pairs but leaves mixed-parity pairs in the conditional sample space. Counting only the even-even and odd-odd pairs would incorrectly exclude six valid even-product outcomes.

Why not the others: (A) uses the unconditional denominator; (C) counts an even first ticket but not the pair condition; (D) assumes only even-even and odd-odd outcomes.

Key point: Rebuild the conditioned sample space.

Q39 / C identical-item grouping

Answer: (C) 15.

Given: Count solutions $n_1+n_2+n_3=4$ with $n_i \geq 0$.

Formula/Principle: Stars and bars gives $C(4+3-1,3-1)$.

Step 1: $C(6,2)=15$.

Worked derivation:

$$N = \binom{6}{2} = \frac{6 \cdot 5}{2} = 15.$$

Why: Identical items within each type make order irrelevant, so permutations of the same count triple are not different choices. Size four already guarantees nonempty.

Why not the others: (A) misses allocations; (B) uses fewer stars; (D) removes one valid triple.

Key point: Count type multiplicities, not sequences.

Q40 / D inverse constant-term coefficient parameter

Answer: (D) 5.

Given: A constant term choosing k factors x^{-2} has coefficient $C(6,k)=15$.

Formula/Principle: The coefficient condition permits $k=2$ or $k=4$; the exponent condition is $p(6-k)-2k=0$.

Step 1: $k=2$ gives $p=1$, while $k=4$ gives $p=4$; add the two admissible positive values.

Worked derivation:

$$\binom{6}{k} = 15 \Rightarrow k \in \{2,4\}; \quad p = \frac{2k}{6-k} \Rightarrow p \in \{1,4\}, \quad \sum p = 5.$$

Why: The same binomial coefficient occurs at two symmetric term indices, so there are two separate exponent equations to check. Selecting only one occurrence of coefficient 15 would miss one valid parameter.

Why not the others: (A) and (B) report individual parameters, while (C) adds the two term indices rather than the parameters.

Key point: Solve coefficient and exponent conditions together.

Q41 / A series convergence under stated condition

Answer: (A) [1,3).

Given: Write $r=x-2$ and distinguish $|r|<1$ from the two endpoints.

Formula/Principle: The series $\sum r^n/n$ converges absolutely for $|r|<1$; at $r=-1$ it is alternating harmonic, while at $r=1$ it is harmonic.

Step 1: $1<x<3$ converges absolutely; $x=1$ converges conditionally; $x=3$ diverges.

Worked derivation:

$$|x-2|<1 \Rightarrow 1<x<3; \qquad x=1: \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \text{ converges}; \quad x=3: \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges}.$$

Why: A radius test determines only the open interval and leaves both boundary points undecided.

The alternating test admits the left boundary but the harmonic comparison excludes the right boundary.

Why not the others: (B) discards a conditionally convergent endpoint; (C) admits a divergent harmonic endpoint; (D) confuses x with the shifted base r.

Key point: Check each endpoint separately.

Q42 / B common-root condition

Answer: (B) 9.

Given: First polynomial factors as (x−1)(x−2), with roots 1 and 2.

Formula/Principle: A common root r satisfies $r^2-kr+4=0$, so $k=r+4/r$.

Step 1: r=1 gives k=5; r=2 gives k=4; sum=9.

Worked derivation:

$$k_1 = 1 + 4 = 5, \quad k_2 = 2 + 2 = 4, \quad k_1 + k_2 = 9.$$

Why: Both roots of the first polynomial can independently be common roots for different parameter values. The question asks for the sum of those possible parameters rather than either single one.

Why not the others: (A) counts only the first possibility; (C) mistakes one parameter as 2; (D) adds neither correct pair.

Key point: Check each candidate common root.

Q43 / C affine variance transform

Answer: (C) 36.

Given: Adding constant 5 does not change deviations; multiplying by 2 doubles each deviation.

Formula/Principle: $\mathrm{Var}(aX+b)=a^2\mathrm{Var}(X)$.

Step 1: $2^2\times 9=36$.

Worked derivation:

$$\mathrm{Var}(2X + 5) = 4\,\mathrm{Var}(X) = 4(9) = 36.$$

Why: Variance averages squared deviations from the transformed mean, so scaling appears squared and translation cancels. Confusing standard deviation with variance would give 18.

Why not the others: (A) adds scales; (B) multiplies by 2 only; (D) squares the translated value.

Key point: Scale variance quadratically.

Q44 / C formation from curve family

Answer: (C) $y'=2y/x$.

Given: Differentiate $y=Cx^2$ to obtain $y'=2Cx$.

Formula/Principle: From original equation $C=y/x^2$.

Step 1: $y'=2(y/x^2)x=2y/x$.

Worked derivation:

$$y' = 2Cx = 2(y/x^2)x = 2y/x.$$

Why: Eliminating the arbitrary constant creates a relation common to the entire family, rather than a solution for one selected C. The x>0 domain makes division by x valid throughout.

Why not the others: (A) retains C; (B) misses factor 2; (D) inverts y.

Key point: Differentiate and substitute the parameter away.

Q45 / A exponential limit

Answer: (A) 2.

Given: e^(2x)=1+2x+(2x)²/2+O(x³).

Formula/Principle: Leading remaining numerator term is 2x².

Step 1: Divide by x² then take x→0.

Worked derivation:

$$\frac{e^{2x}-1-2x}{x^2}=2+O(x)\longrightarrow 2.$$

Why: Subtracting both constant and linear terms isolates the quadratic contribution; direct substitution gives an indeterminate zero over zero. The limit is finite because higher powers vanish after division.

Why not the others: (B) drops factor 2; (C) forgets factorial; (D) substitutes directly.

Key point: Expand to the first surviving order.

Physics

Q46 / B vertical-motion turning time

Answer: (B) 2 s.

Given: v(t)=u−gt.

Formula/Principle: At a smooth turning point vertical velocity is zero.

Step 1: 20−10t=0, so t=2 s.

Worked derivation:

$$t_{\text{top}}=u/g=20/10=2\ \text{s}.$$

Why: The balcony elevation changes the absolute height but not the time for upward velocity to reach zero. Gravity acts downward during the entire flight, including the turning instant.

Why not the others: (A) leaves positive speed; (C) and (D) occur after turning.

Key point: Solve velocity zero, not displacement zero.

Q47 / C coupled tension with support acceleration

Answer: (C) 36 N.

Given: Mass has upward acceleration 2 relative to ground.

Formula/Principle: T−mg=ma.

Step 1: T=3(10+2)=36 N.

Worked derivation:

$$T=m(g+a)=3(12)=36\ \text{N}.$$

Why: Stationary relative to the accelerating cabin does not mean zero acceleration in an inertial frame. The cord must both balance gravity and supply net upward force.

Why not the others: (A) uses g−a; (B) ignores acceleration; (D) overadds it.

Key point: Specify the reference frame for acceleration.

Q48 / D position-dependent work

Answer: (D) 2√3 m s⁻¹.

Given: Initial kinetic energy is ½(2)(2²)=4 J; the two graph areas are 4 J each.

Formula/Principle: Work–energy theorem Kf=Ki+∫F dx.

Step 1: Total work 8 J gives Kf=12 J and ½(2)v²=12.

Worked derivation:

$$W = \int_0^2 2x\,dx + \int_2^3 4\,dx = 8\ \text{J}, \quad v = \sqrt{\frac{2(4+8)}{2}} = 2\sqrt{3}\ \text{m s}^{-1}.$$

Why: The rising and constant segments contribute distinct areas that must be added before applying the work–energy theorem. The initial kinetic energy remains part of the final balance; setting final energy equal to work alone loses the starting speed.

Why not the others: (A) ignores work; (B) equates final kinetic energy to work alone; (C) treats energy increase as an increase in speed.

Key point: Integrate force, then add initial energy.

Q49 / A rolling energy partition

Answer: (A) 1/3.

Given: I=½mR² and ω=v/R.

Formula/Principle: Krot=½Iω²=¼mv²; Ktrans=½mv².

Step 1: Krot/Ktotal=(1/4)/(3/4)=1/3.

Worked derivation:

$$\frac{K_{\text{rot}}}{K_{\text{total}}} = \frac{\frac{1}{4}mv^2}{\frac{3}{4}mv^2} = \frac{1}{3}.$$

Why: Static friction allows rolling without dissipation, but kinetic energy splits between translation and rotation. Using mgh alone without partition would miss the required fraction.

Why not the others: (B) uses I ratio directly; (C) ignores rotational energy; (D) reverses the partition.

Key point: Combine the two kinetic contributions.

Q50 / B escape speed ratio

Answer: (B) 1.

Given: vesc=√(2GM/R).

Formula/Principle: Ratio √[(2M/2R)/(M/R)].

Step 1: Ratio=1.

Worked derivation:

$$\frac{v_{e,p}}{v_{e,E}} = \sqrt{\frac{M_p/R_p}{M_E/R_E}} = \sqrt{\frac{2}{2}} = 1.$$

Why: Doubling gravitational mass strengthens attraction while doubling launch radius weakens it by the same factor in the energy ratio. Surface gravity itself need not stay equal.

Why not the others: (A) scales radius alone; (C) mass alone; (D) omits square root.

Key point: Escape speed scales with √(M/R).

Q51 / C buoyancy under partial immersion

Answer: (C) 0.6.

Given: Weight=ρblock gV; buoyancy=ρwater gVsub.

Formula/Principle: Floating balance gives Vsub/V=ρblock/ρwater.

Step 1: 600/1000=0.6; uniform cross section gives same height fraction.

Worked derivation:

$$h_{\text{sub}}/h = V_{\text{sub}}/V = 600/1000 = 0.6.$$

Why: The fluid displaces only the submerged volume, not the whole block. The density ratio determines immersion for a freely floating object before any external force is added.

Why not the others: (A) gives emergent fraction; (B) assumes half; (D) implies neutral or denser block.

Key point: Balance displaced-water weight with block weight.

Q52 / D first-law sign convention

Answer: (D) +30 J.

Worked derivation:

$$\Delta U = Q - W = 50\ \text{J} - 20\ \text{J} = +30\ \text{J}.$$

Why: Heat entering the gas increases internal energy, while work done by the gas transfers part of that input out. The explicit work sign convention prevents the opposite sign choice.

Why not the others: (A) reverses both terms; (B) reverses net sign; (C) adds rather than subtracts work.

Key point: Define work sign before applying first law.

Q53 / A field cancellation locus

Answer: (A) Midpoint.

Worked derivation:

$$E_{-a}(0) = \frac{kq}{a^2}\hat{x}, \quad E_{+a}(0) = -\frac{kq}{a^2}\hat{x} \Rightarrow E_{\text{net}}(0) = 0.$$

Why: Between like charges the fields point oppositely and equal magnitudes require equal distances, hence the midpoint. Outside the two charges both fields point the same way and cannot cancel.

Why not the others: (B) and (C) lie outside with co-directed fields; (D) overlooks symmetry.

Key point: Check direction before equating magnitudes.

Q54 / B dielectric with battery disconnected

Answer: (B) 1/4.

Given: Disconnection fixes Q; dielectric raises C to 4C.

Formula/Principle: U=Q²/(2C).

Step 1: Uf/Ui=C/Cf=1/4.

Worked derivation:

$$\frac{U_f}{U_i} = \frac{Q^2/(2 \cdot 4C)}{Q^2/(2C)} = \frac{1}{4}.$$

Why: With no battery pathway, charge cannot adjust while dielectric polarisation raises capacitance. The lost field energy corresponds to work associated with dielectric insertion, so fixed-voltage reasoning would answer another setup.

Why not the others: (A) reverses ratio; (C) squares it; (D) uses C².

Key point: Identify isolated versus battery-connected boundary.

Q55 / C force with vector field components

Answer: (C) +z, 6 N.

Given: v perpendicular to B.

Formula/Principle: F=qv×B, magnitude qvB sin90°.

Step 1: i×j=k, 1×2×3=6 N.

Worked derivation:

$$\mathbf{F} = (1)(2\mathbf{i}) \times (3\mathbf{j}) = 6\mathbf{k}\ \text{N}.$$

Why: The magnetic force is perpendicular to both vectors and the positive charge follows the right-hand orientation. Reversing charge sign or vector order would reverse direction without changing magnitude.

Why not the others: (A) reverses sign; (B) misses speed; (D) chooses field direction.

Key point: Cross-product order fixes force sign.

Q56 / D inductor energy from current

Answer: (D) 9 J.

Given: L=2 H, I=3 A.

Formula/Principle: U=½LI².

Step 1: ½×2×9=9 J.

Worked derivation:

$$U = \frac{1}{2}(2)(3)^2 = 9\ \text{J}.$$

Why: Current appears squared because building the field requires work against a growing self-induced emf. Inductance alone or current alone does not determine energy.

Why not the others: (A) omits square; (B) omits half; (C) mishandles both.

Key point: Square current before multiplication.

Q57 / A real-virtual image sign

Answer: (A) Virtual upright at 10 cm left.

Given: f=+10 cm, u=−5 cm.

Formula/Principle: 1/f=1/v−1/u.

Step 1: 1/v=1/10−1/5=−1/10, so v=−10 cm; magnification v/u=+2.

Worked derivation:

$$v = -10\ \text{cm}, \quad m = v/u = (-10)/(-5) = +2.$$

Why: An object inside the focal length produces outgoing diverging rays whose backward extensions meet on the object side. A negative image distance and positive magnification together identify virtual upright imaging.

Why not the others: (B) reverses distance sign; (C) mismatches sign and orientation; (D) calls a negative-v image real.

Key point: Interpret both v sign and magnification.

Q58 / B diffraction central width

Answer: (B) Doubles.

Given: First minima approximately at sinθ=±λ/a.

Formula/Principle: Central linear width≈2Dλ/a at small angles.

Step 1: Replace a with a/2; width doubles.

Worked derivation:

$$w' = \frac{2D\lambda}{a/2} = 2w.$$

Why: Narrowing the aperture spreads the diffraction pattern because the first zero shifts farther from centre in angle. The two sides of central lobe both scale the same way.

Why not the others: (A) treats width proportional to a; (C) squares effect; (D) ignores slit.

Key point: Diffraction width is inverse to slit width.

Q59 / C line-series limit

Answer: (C) Discrete emission lines.

Worked derivation:

$$\Delta E = h\nu = hc/\lambda \quad \text{for allowed atomic transitions.}$$

Why: Isolated excited atoms emit photons at differences between their allowed energy levels, yielding characteristic lines. A dense thermal source blends many processes into a more continuous spectrum under the usual conditions.

Why not the others: (A) ignores discrete levels; (B) denies emission; (D) ignores element-specific transitions.

Key point: Relate line positions to atomic energy gaps.

Q60 / D radioactivity count rate

Answer: (D) 275 per minute.

Given: Source-only initial rate=800−100=700.

Formula/Principle: Two half-lives reduce source rate to 700/4=175; background remains 100.

Step 1: Total=175+100=275 min^{-1}.

Worked derivation:

$$R(t) = 100 + (800 - 100)2^{-2} = 100 + 175 = 275 \text{ min}^{-1}.$$

Why: Background counts do not decay with the source in the stipulated measurement, so applying the factor one quarter to the observed total would underpredict the final reading. Subtract and restore the constant background around the decay operation.

Why not the others: (A) omits background; (B) quarters total; (C) adds the initial background twice.

Key point: Separate source signal from baseline count.
